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Project Title : Securing Stigmergic Communication in Cooperative Multi-Agent Reinforcement Learning Swarms

Capstone Project Phase-1
End Semester Assessment

Project ID : PW26_BJD_21
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Problem: Cooperative Resource Retrieval

- Several agents; one shared resource-retrieval task
- Each agent senses only a nearby 3 × 3 region
- Food and home pheromone trails support indirect communication
- Shared PPO chooses actions from each agent’s local view

Explore → Collect food → Return to nest

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Problem: Misleading Shared Signals

- One compromised agent can add false food pheromone
- Shared trails reveal concentration, not their author
- Threat: nonnegative deposits at occupied cells; fixed budgets
- Defense goal: local history and own progress; preserve clean retrieval

Research question: Can local evidence limit disruption?

Literature Findings and the Candidate Gap

Area / core papers	Established finding	Implication for this project
Pheromone attacks [1]	False trails and cautionary defense	Injection itself is prior work
Learned coordination [2,6]	Environmental traces can support learning	Verify benefit for our PPO setup
Field / sensing limits [4,5]	Dynamics and density affect signals	Measure visibility and clean cost
MARL security / PPO [3,7,8]	Attack access and evaluation matter	Matched controls; no robustness assumption

Candidate gap: local defense for learned coordination under controlled shifts

Focused comparison: 8 core papers | Broader 30+ survey still in progress

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Applications and Domains

Potential domain	Coordination need	Possible benefit to investigate
Warehouse retrieval	Several agents collect and return items	Less wasted travel from misleading cues
Exploration and disaster search	Local decisions with limited communication	More dependable use of shared traces
Distributed inspection	Repeated visits across a shared area	Recognise unproductive revisitation

Current evidence: simulated grid retrieval only

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Review Inputs and Our Response

Review input	Action taken	Current position
How will the dataset be built?	Matched triplets → corpus → local CSV	Development archive; labels/splits pending
Implementation is complicated	Phase 2A / 2B; module and integration checks	172 tests passed in recorded Nov 6 check
Show useful progress and new code	Fair trained baselines; attack diagnostics	Clean benefit; variable attack damage
Strengthen the literature / gap	Area-wise critical comparison	8 core papers; wider survey incomplete

Overall Architecture and Next Work

Versioned simulator
Grid • seeds • food/home fields

→

Clean training
Local 58 values → shared PPO

↓ freeze checkpoint

Matched frozen-policy rollouts
Clean / disabled / attacked
Bounded occupied-cell injector

←

Separated records
Local observations / actions / history
Audit: identities / injection / totals

↓

Corpus builder → CSV exporter
58 observations + 12 history features
Metadata / outcomes kept separate

→

Planned detector and mitigation
Labels → grouped splits → windows
Required baselines + held-out tests

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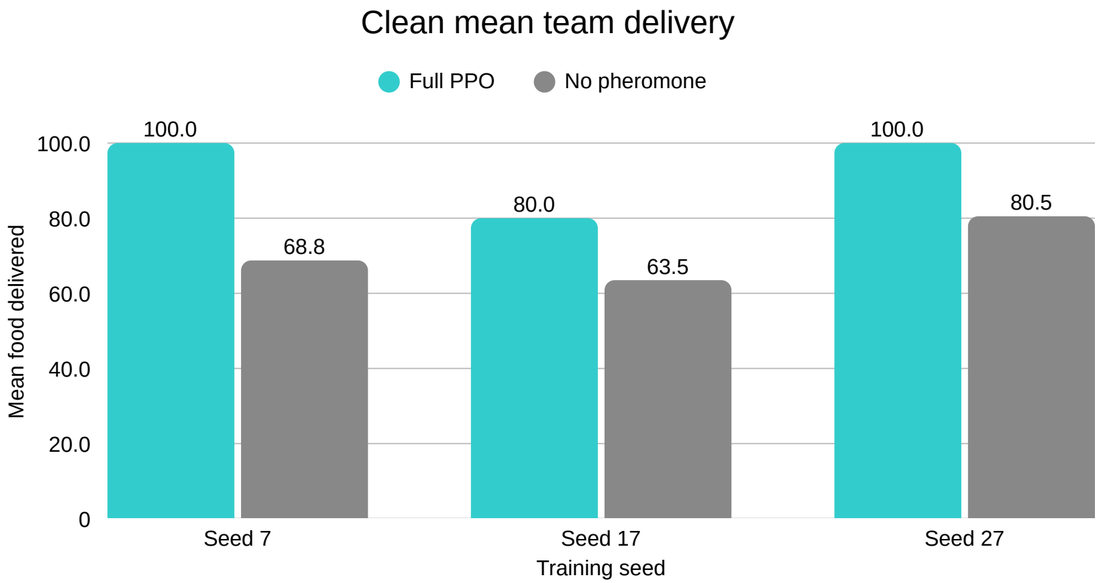
Dataset Strategy: Candidates and Final Status

Candidate / selected source	What it can provide	Decision
External swarm / MARL resources	Related tasks and methodology	Context; not a matching ready-made dataset
Our simulator-generated triplets	Matched local trajectories + separate audit	Selected data-generation strategy
Historical development corpus	36 groups / 108 episodes / 412,016 rows	Archived development evidence only
Final detector dataset	Exposure labels / grouped partitions / windows	Not finalised; detector training pending

PPO: 58 inputs | CSV candidates: 58 + 12 | New 432 diagnostic episodes remain separate

Implementation and Current Evidence

Implemented: fair PPO baselines • matched runs • corpus/CSV • input-response diagnostics



Clean benefit reproduced on 3 seeds
Deterministic full: 100 / 0 / 60



Loss: 10.53 units (~15%); only 18/36 pairs damaged
3/step, cap 3,300; consistency gate still fails

172 tests passed (6 Nov) | No trained detector, learned defense or held-out result

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Thank You

Questions

PW26_BJD_21 | Phase-1 ESA | 16 November 2026

Backup: Learned Stigmergy and Field Mechanics

Paper	Contribution	Boundary for our study
Xu et al. [2], 2022	Learned digital-pheromone collaboration	Independent/federated learning; not our shared PPO
Aina and Ha [6], 2026	Stigmergy + curriculum for coordination	Congestion task/controller differs
Hunt et al. [4], 2019	Density-related limits of pheromone use	Benign exploration; not an injection dataset
Salman et al. [5], 2020	Phormica artificial-pheromone sensing	Physical inspiration; no hardware claim here

Backup: Security and Learning Methods

Paper	Contribution	Boundary for our study
Aswale et al. [1], 2022	Misleading pheromone + cautionary mitigation	Attack and mitigation are prior work
Standen et al. [7], 2025	MARL attack/defense taxonomy	Survey; not our trajectory source
Li et al. [8], 2025	Adversarial minority influence	Learned action-based attack; different access
Schulman et al. [3], 2017	PPO policy optimisation	Learning method; no swarm-security guarantee

Backup: Candidate J and Fair Comparisons

12 × 12 • 4 agents • 100 food • 1,100 steps • 2% evaporation

Training seed	Full PPO	No pheromone	Random	Full – no pheromone
7	100.00	68.75	3.00	+31.25
17	80.00	63.50	1.75	+16.50
27	100.00	80.50	2.25	+19.50

Six fresh PPO jobs: 2 input modes × 3 seeds
300,000 requested / 300,032 actual agent transitions per job
Stochastic development gate passed; deterministic full means: 100 / 0 / 60

Backup: All Attack Outcomes

Setting	Request / cap / onset	Mean honest loss	Positive pairs
Historical immediate	1 / 20 / first transition	−2.17	1 / 12
Historical longer	1 / 200 / first transition	−0.42	2 / 12
Historical delayed	1 / 200 / transition 201	−2.25	1 / 12
Sustained one	1 / 1,100 / first transition	+2.47	13 / 36
Sustained three	3 / 3,300 / first transition	+10.53	18 / 36
Local east-edge decoy	3 / 3,300 / altered routing	−1.47	9 / 36

Loss = honest(disabled) − honest(attacked) | All consistency gates fail

Backup: Data Boundary and Remaining Work

Recorded field family	Columns	Use
Local observation + own history	58 + 12	Candidate detector inputs; PPO still uses58
Run / group / seed provenance	10	Grouping and audit only
Actions / reward / transition outcomes	9	Separate from default predictors
Privileged simulator metadata	Separate audit files	Later label construction; never live inputs

Next: confirm attack → validate exposure labels → grouped partitions → windows
Then: local detector / mitigation → required baselines → held-out evaluation

Backup: References 1

[1] A. Aswale, A. Lopez, A. Ammartayakun and C. Pinciroli, “Hacking the Colony: On the Disruptive Effect of Misleading Pheromone and How to Defend Against It,” AAMAS, pp.27–34,2022. https://arxiv.org/abs/2202.01808

[2] X. Xu, R. Li, Z. Zhao and H. Zhang, “Stigmergic Independent Reinforcement Learning for Multiagent Collaboration,” IEEE TNNLS, vol.33,no.9,pp.4285–4299,2022. https://doi.org/10.1109/TNNLS.2021.3056418

[3] J. Schulman et al., “Proximal Policy Optimization Algorithms,” arXiv:1707.06347,2017. https://arxiv.org/abs/1707.06347

[4] E. R. Hunt, S. Jones and S. Hauert, “Testing the limits of pheromone stigmergy in high-density robot swarms,” R. Soc. Open Sci.,6(11),190225,2019. https://doi.org/10.1098/rsos.190225

Backup: References 2

[5] M. Salman, D. Garzon Ramos, K. Hasselmann and M. Birattari, “Phormica: Photochromic Pheromone Release and Detection System for Stigmergic Coordination in Robot Swarms,” Front. Robot. AI,7,591402,2020. https://doi.org/10.3389/frobt.2020.591402

[6] K. O. Aina and S. Ha, “Deep reinforcement learning for multi-agent coordination,” Artif. Life Robot.,31,pp.484–494,2026. https://doi.org/10.1007/s10015-025-01089-z

[7] M. Standen, J. Kim and C. Szabo, “Adversarial Machine Learning Attacks and Defences in Multi-Agent Reinforcement Learning,” ACM Comput. Surv.,2025. https://doi.org/10.1145/3708320

[8] S. Li et al., “Attacking cooperative multi-agent reinforcement learning by adversarial minority influence,” Neural Netw.,191,107747,2025. https://doi.org/10.1016/j.neunet.2025.107747